

Introduction to Distributed Systems

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What are Systems and Distributed Systems?

Non-Functional System Requirements

Non-Functional System Requirements

- Reliability

Non-Functional System Requirements

- Reliability
- Performance

Non-Functional System Requirements

- Reliability
- Performance
- Scalability

Scaling to Higher Load

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- Vertical Scaling

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Architecture: Shared-Memory

Scaling to Higher Load

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Architecture: Shared-Memory

Drawback: $Cost(2R) > 2Cost(R)$

$Performance(2x) < 2Performance(x)$

Scaling to Higher Load

- Vertical Scaling

Architecture: Shared-Memory

Drawback: $Cost(2R) > 2Cost(R)$

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- Horizontal Scaling

Scaling to Higher Load

- Vertical Scaling

Architecture: Shared-Memory

Drawback: $Cost(2R) > 2Cost(R)$

$Performance(2x) < 2Performance(x)$

- Horizontal Scaling

Architecture: Shared-Nothing

Scaling to Higher Load

- Vertical Scaling

Architecture: Shared-Memory

Drawback: $Cost(2R) > 2Cost(R)$

$Performance(2x) < 2Performance(x)$

- Horizontal Scaling

Architecture: Shared-Nothing

Drawback: Additional Complexity & Operational Overhead

Scalability - Problems

- CPU & Memory Usage

Scalability - Problems

- CPU & Memory Usage
- Storage Read Load

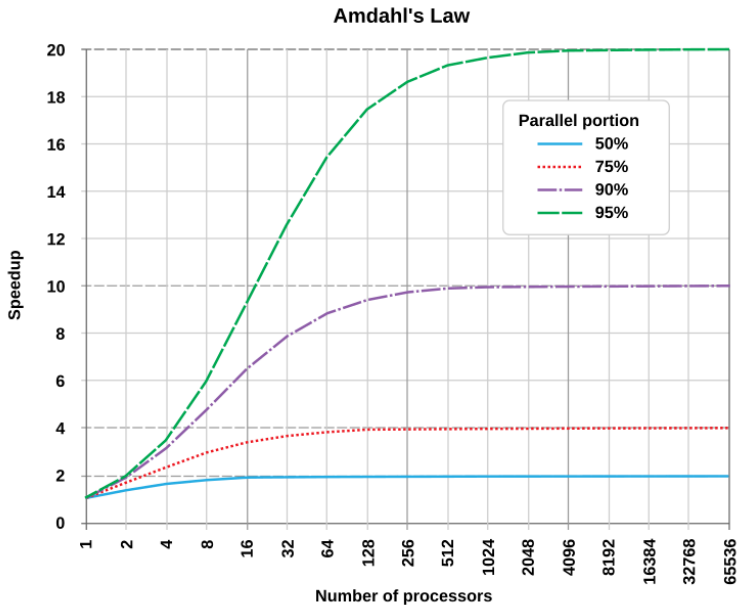
Scalability - Problems

- CPU & Memory Usage
- Storage Read Load
- Storage Data Volume

Scalability - Problems

- CPU & Memory Usage
- Storage Read Load
- Storage Data Volume
- Storage Write Load

Scalability - Problems



Reliability and Performance - Problems

- **Reliability:** System Faults

Reliability and Performance - Problems

- **Reliability:** System Faults
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Reliability and Performance - Problems

- **Reliability:** System Faults
 - Server Outages
 - Network Problems

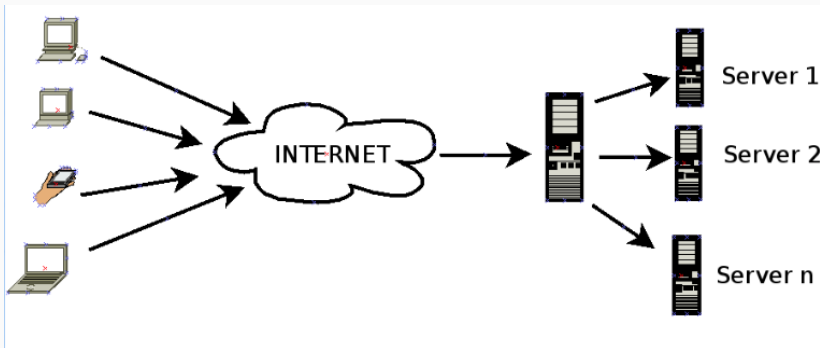
Reliability and Performance - Problems

- **Reliability:** System Faults
 - Server Outages
 - Network Problems
 - Datacenter Outages

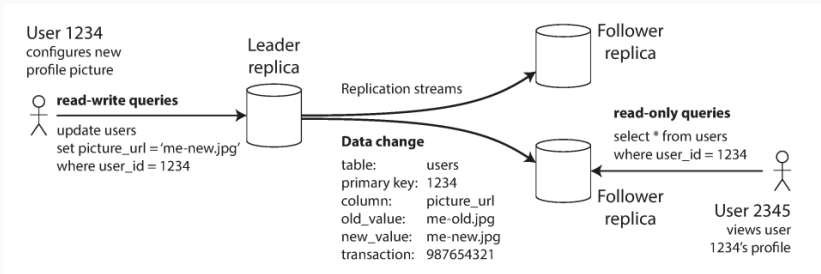
Reliability and Performance - Problems

- **Reliability:** System Faults
 - Server Outages
 - Network Problems
 - Datacenter Outages
- **Performance:** Latency

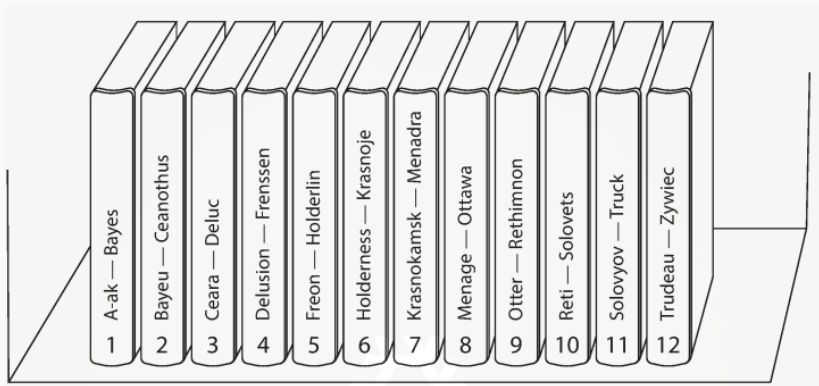
Load Balancing



Replication



Sharding



Scalability - Problems vs Solutions

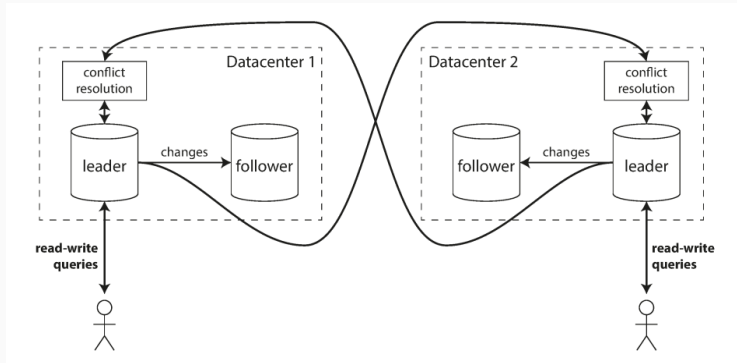
Scalability - Problems vs Solutions

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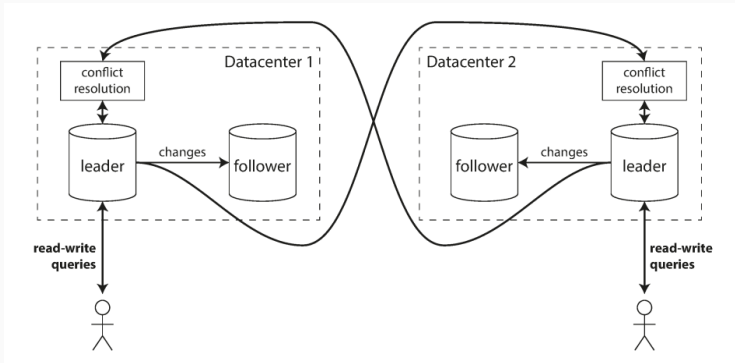
Scalability - Problems vs Solutions

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- Solutions
 - Load Balancing
 - Replication
 - Sharding

Reliability and Performance - Solutions

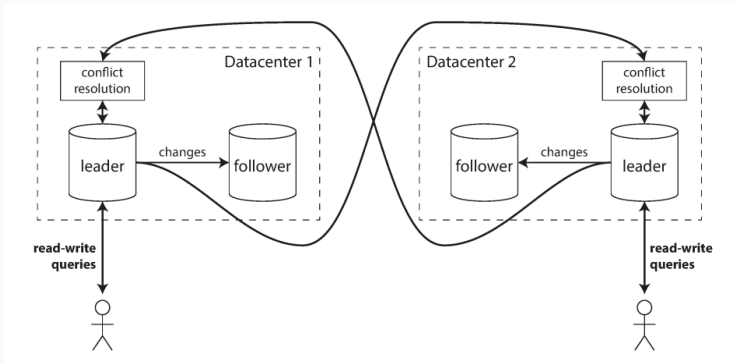


Reliability and Performance - Solutions



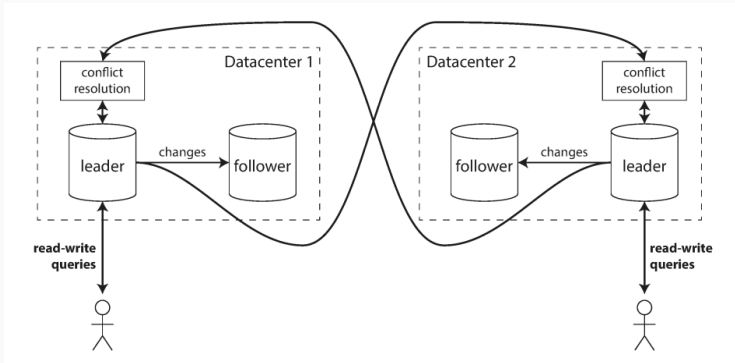
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Reliability and Performance - Solutions



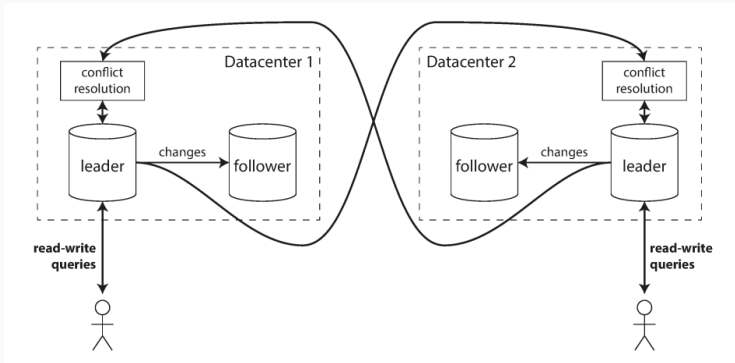
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Reliability and Performance - Solutions



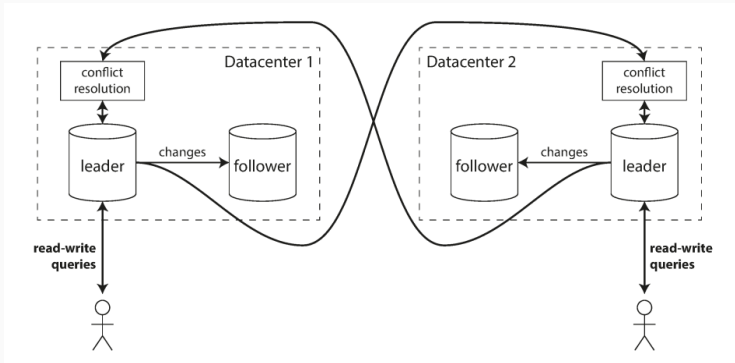
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Reliability and Performance - Solutions



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Reliability and Performance - Solutions



- **Reliability:** System Faults
 - Server Outages
 - Network Problems
 - Datacenter Outages
- **Performance:** Latency

There is no such thing as a
generic, one-size-fits-all
scalable architecture.

Q&A